

SS928V100 平台旷视算法适配指南



版本历史

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目录

1. 文件说明	4
2. 上手流程	4
3. Demo 流程	4
3.1. Demo 编译	4
3.2. 板端授权	5
3.3. Demo 运行	7
4. MPP 适配	9
5. 用户授权	11

1. 文件说明

3402 发布.zip

650.Hisi3402_COMC.870.990.V2.0.0.sdk200.UserPackage.tar //用户包，模型 pkg

650.脸人全量_安防 DATA 【V2.0.0】.zip //data 数据包，图片文件

Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit.tar //SDK 开发包

HISI3402_Store_softlicence 【V2.0.0】.tar.gz //授权工具

HISI_3402_3403 交付文档汇总[V2.0.1].zip //旷视相关 SDK API 说明文档

SDK 交付入门指南.zip //SDK 入门文档

MegSDK-General-AlgoProP-CAP 开发 20250818.cert //通信认证文件，旷视后台生成，以实际文件为准

610.Hisi3402_COMC.870.990.V2.0.0.sdk200.UserPackage.pkg //用户包，结构化模型 pkg

2. 上手流程

- 1.从《SDK 交付入门指南.zip》入手，先熟悉旷视 SDK 说明文档以及相关视频资源，对开发流程有基本的认知。
- 2.熟悉 SS928V100 平台旷视 SDK 文件组成《3402 发布.zip》。
- 3.根据文档资料《HISI_3402_3403 交付文档汇总[V2.0.1].zip》进行算法适配开发。

3. Demo 流程

3.1. Demo 编译

```
#解压 Demo 代码

tar -xvf Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit.tar


#执行脚本编译 Demo

cd Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit

./build.sh


#完成编译后生成的程序位于 bin 目录
```

```
lmq@compilesrv:~/Projects/HiEuler_PI/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit$
lmq@compilesrv:~/Projects/HiEuler_PI/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit$ ls bin/
algo_alarm  common_object_image_recg  facehuman_attr  face_search  face_video_search  hvccfp_attr  sdk_performance  test_tar_install_uninstall
common_image  demo_capability  facehuman_video_capture  face_verify  human_video_capture  hvccfp_video_capture  test_auth
lmq@compilesrv:~/Projects/HiEuler_PI/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit$
```

3.2. 板端授权

(1) 获取授权信息，以下仅为示例，实际以旷视算法账号管理员提供型号为准。

授权SKU	授权名称	版本	授权类型	适配设备型号	授权计量单位	状态	详情	有效期	操作
02410189	MegSDK-General-Alg...	2.0.0.Hisi...	开发	Hisi3402_COMC.870.990	按路	正常	查看	1月	绑定下载 资产明细 预警提醒配置

02410189

MegSDK-General-AlgoProP-CAP

2.0.0.Hisi3402 开发

Hisi3402_COMC.870.990

(2) 生成指纹文件

#板端执行

指定授权文件目录(必须是用读写权限的空文件夹):

```
export HASPUSER_PREFIX=/root/megvii
```

生成 finger 文件:

```
./three_step_auth_tool_for_HISI3402 -t Hisi3402_COMC.870.990 -i enb340320250822
```

-t 为适配设备型号

-i 为用户自定义字段, 可以是设备 SN、MAC、IP 等

给予权限, 方便服务器生成设备授权文件

```
chmod 777 enb340320250822.finger
```

```
/mnt/megvii #  
/mnt/megvii # export HASPUSER_PREFIX=/root/megvii  
/mnt/megvii #  
/mnt/megvii # ./three_step_auth_tool_for_HISI3402 -t Hisi3402_COMC.870.990 -i en  
b340320250822  
platform is hisi3402 !!!  
exit status value = [0x0]  
run shell script successfully.  
exit status value = [0x0]  
run shell script successfully.  
save hisi3402 fingerprint sucess!!!  
/mnt/megvii #
```

(3) 生成设备授权文件

#服务器执行:

单一设备授权执行:

```
./three_step_auth_tool_for_X86_64 -f ./enb340320250822.finger  
-c ./MegSDK-General-AlgoProP-CAP20250818.cert
```

运行生成后生成的 lic 文件位于压缩包 tar 中。

```
lmq@compilesrver:~/nfs/SS928V100/megvii$ ./three_step_auth_tool_for_X86_64 -f ./enb340320250822.finger -c ./MegSDK-General-AlgoProP-CAP20250818.cert  
save lic tar success!!  
lmq@compilesrver:~/nfs/SS928V100/megvii$  
lmq@compilesrver:~/nfs/SS928V100/megvii$ ls  
1755832787.tar demo_capability.sh hisilicon MegSDK-General-AlgoProP-CAP20250818.cert run.sh three_step_auth_tool_for_HISI3402  
bin enb340320250822.finger lib models stub three_step_auth_tool_for_X86_64  
lmq@compilesrver:~/nfs/SS928V100/megvii$
```

(4) 设备授权

#板端执行:

```
./three_step_auth_tool_for_HISI3402 -v ./enb340320250822_2e58af34-8193-495e-b1d8-04470024a888.lic
```

执行成功后, 授权会写入 HASPUSER_PREFIX 环境变量指向的目录。

#注意: 如果重复运行时, 需手动设置环境变量

```
export HASPUSER_PREFIX=/root/megvii
```

```
/mnt/megvii # ./three_step_auth_tool_for_HISI3402 -v ./enb340320250822_2e58af34-8193-495e-b1d8-04470024a888.lic
exit status value = [0x0]
run shell script successfully.
exit status value = [0x0]
run shell script successfully.
exit status value = [0x0]
run shell script successfully.
auth active success!!!
/mnt/megvii #
```

(5) 查看板端授权信息

```
./three_step_auth_tool_for_HISI3402 -ll
```

#注意: 需留意 algo index 后面的授权模型信息<0,610.Hisi3402_COMC.870.990>, 后续运行程序时模型文件需与这个匹配。

#注意需先设备板端时间

```
date -s "2025-8-27 11:20"
```

```
/mnt/megvii # ./three_step_auth_tool_for_HISI3402 -ll
auth file name:enb340320250822_2e58af34-8193-495e-b1d8-04470024a888.lic
****display auth info****
auth file uuid:2e58af34-8193-495e-b1d8-04470024a888
algos:
algo index:0,610.Hisi3402_COMC.870.990
algo cards:
algo cards index:0,Algo-1
Auth OK!!!
****display auth info****
/mnt/megvii #
```

3.3. Demo 运行

#注意: Demo 运行必须在原工程路径, NPU 库依赖于工程路径, 否则会找不到相关库, 如下图所示。

```
/mnt/megvii # ./demo_capability.sh
[EVENT] CCECPU(1843,demo_capability):2025-08-22-15:44:43.782.200 [aicpusd_cust_so_manager.cpp:73][AicpuCustSoManagerInit][tid:1843] cust so dir name is /root/cust_aicpu_0_1843/.
[ERROR] stub library cannot be used for execution, please check your environment variables and compilation options to make sure you use the correct ACL library.
[2025/08/22 15:44:43][ERROR][MgvdDeviceInit:404][svp_acl_init] fail: (100039).
[2025/08/22 15:44:43][ERROR][MgvdDeviceInit:405]error at /work/xueyangliu/algstore/ss927/megsdk-pipe/hal/impl/device/ss927/ss927_device.cpp:405 MgvdDeviceInit
exit status value = [0x0]
run shell script successfully.
decrypt3 with mbedtls
INFO: main(262): sdk version : Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit
facehuman pkg support 7 demos
facehuman demo index = 0,name = face_search
facehuman demo index = 1,name = face_verify
facehuman demo index = 2,name = face_video_search
facehuman demo index = 3,name = facehuman_attr
facehuman demo index = 4,name = facehuman_performance
facehuman demo index = 5,name = facehuman_video_capture
facehuman demo index = 6,name = facehuman_image_detect
-----
hvcfp pkg support 0 demos
-----
other pkg support 0 demos
-----
one obj feature pkg support 0 demos
[ERROR] stub library cannot be used for execution, please check your environment variables and compilation options to make sure you use the correct ACL library.
[2025/08/22 15:44:43][ERROR][MgvdDeviceUnit:432][svp_acl_finalize] fail: (100039).
[EVENT] CCECPU(1843,demo_capability):2025-08-22-15:44:44.075.014 [aicpusd_mpi_mgr.cpp:49][PrintStatisticInfo][tid:1843] Mpi Dvpp event statistic: [0]
/mnt/megvii #
```

(1) 将代码工程通过 NFS 挂载到板端。

```
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit #
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit # ls
3rd_sdk      bin          build.sh     demo_capability.sh  lib          output       sample
auth_file    build        data         include            models      run.sh
```

(2) 运行 demo_capability.sh 获取 SDK 能力集，并生成 run.sh 脚本。

#注意：获取 SDK 能力集前请确保 models 目录下的模型文件与授权文件匹配，且只有一个模型文件。

```
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit #
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit # ls models/
610.Hisi3402_COMC.870.990.V2.0.0.sdk200.UserPackage  610.Hisi3402_COMC.870.990.V2.0.0.sdk200.UserPackage.pkg
```

```
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit # ./demo_capability.sh
[EVENT] CCECPU(2058,demo_capability):2025-08-27-15:59:04.439.970 [aicpusd_cust_so_manager.cpp:73][AicpuCustSoManagerInit][tid:2058] cust so dir name is /root/cust_aicpu_0_2058/.
[EVENT] PROFILING(2058,demo_capability):2025-08-27-15:59:04.522.347 [msprof_callback_impl.cpp:99] >>> (tid:2058) MsprofCtrlcallback called, type: 255
[EVENT] PROFILING(2058,demo_capability):2025-08-27-15:59:04.522.684 [prof_acl_mgr.cpp:1333] >>> (tid:2058) Init profiling for dynamic profiling
exit status value = [0x0]
run shell script successfully.
decrypt3 with mbedtls
INFO: main(262): sdk version : Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit
facehuman pkg support 0 demos
-----
hvcfp pkg support 6 demos
hvcfp demo index = 0,name = hvcfp_attr
hvcfp demo index = 1,name = hvcfp_performance
hvcfp demo index = 2,name = hvcfp_video_capture
hvcfp demo index = 3,name = hvcfp_image_attr
hvcfp demo index = 4,name = hvcfp_image_attr_all
hvcfp demo index = 5,name = hvcfp_image_detect
-----
other pkg support 0 demos
-----
one obj feature pkg support 0 demos
[EVENT] ASCENDCL(2058,demo_capability):2025-08-27-15:59:04.549.336 [resource_statistics.cpp:104]2058 TraverseStatistics: The ResourceType:ACL_STATISTICS_MALLOC_FREE, applyTotal = 0, a
Total = 0, releaseSucc = 0
```

(3) 运行 run.sh 执行对应的 demo 程序，程序运行结果位于 output 目录下。

```

/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit # ./run.sh
##### help:
##### ./run.sh [args1]
##### args 1: hvcfp_attr hvcfp_performance hvcfp_video_capture hvcfp_image_attr hvcfp_image_attr_all hvcfp_image_detect
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit # ./run.sh hvcfp_attr
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:50.630.451 [aicpusd_cust_so_manager.cpp:73][AicpuCustSoManagerInit][tid:2162] cust
[EVENT] PROFILING(2162,hvcfp_attr):2025-08-27-19:44:50.712.345 [msprof_callback_impl.cpp:99] >>> (tid:2162) MsprofCtrlCallback call
[EVENT] PROFILING(2162,hvcfp_attr):2025-08-27-19:44:50.712.684 [prof_acl_mgr.cpp:1333] >>> (tid:2162) Init profiling for dynamic pr
exit status value = [0x0]
run shell script successfully.
decrypt3 with mbedtls
INFO: main(90): sdk version : Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit
create ../models//610.Hisi3402_COMC.870.990.V2.0.0.sdk200.UserPackage success!
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.099.691 [ae_so_manager.cc:440][LoadSo][tid:2162][AICPU_PROCESSER] Single so mar
ce/libaicpu_kernels.so.
[ERROR] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.099.877 [ae_so_manager.cc:453][LoadSo][tid:2162][AICPU_PROCESSER] Single so mar
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.099.939 [ae_kernel_lib_aicpu.cc:168][BatchLoadKernelSo][tid:2162][AICPU_PROCESS
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.099.997 [aicpusd_interface.cpp:114][AICPUPreOpenKernels][tid:2162] Preload so l
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.100.082 [ae_so_manager.cc:440][LoadSo][tid:2162][AICPU_PROCESSER] Single so mar
ce/libcpu_kernels.so.
[ERROR] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.100.150 [ae_so_manager.cc:453][LoadSo][tid:2162][AICPU_PROCESSER] Single so mar
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.100.206 [ae_kernel_lib_aicpu.cc:168][BatchLoadKernelSo][tid:2162][AICPU_PROCESS
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:44:52.100.260 [aicpusd_interface.cpp:114][AICPUPreOpenKernels][tid:2162] Preload so l
[EVENT] PROFILING(2162,hvcfp_attr):2025-08-27-19:44:52.103.214 [msprof_callback_impl.cpp:120] >>> (tid:2162) MsprofSetDeviceCallbac
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:52.242.822 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
]
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:52.597.639 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:52.651.583 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:52.729.564 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:52.765.600 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:52.810.589 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:53.014.959 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
[EVENT] GE(2162,hvcfp_attr):2025-08-27-19:44:53.135.801 [davinci_model.cc:345]2162 InitFeatureMapAndP2PMem:[IMAS]InitFeatureMapAndP
[EVENT] ASCENDE(2162,hvcfp_attr):2025-08-27-19:45:04.801.398 [resource_statistics.cpp:95]2162 TraversStatistics: The ResourceType:39 is not exist.
[EVENT] PROFILING(2162,hvcfp_attr):2025-08-27-19:45:04.861.454 [msprof_callback_impl.cpp:99] >>> (tid:2162) MsprofCtrlCallback called, type: 3
[EVENT] CCECPU(2162,hvcfp_attr):2025-08-27-19:45:05.053.140 [aicpusd_mpi_mgr.cpp:49][PrintStatisticInfo][tid:2162] Mpi Dvpp event statistic: [0]
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit #
body cycle draw frame face plate vehicle
/mnt/megvii/Hisi3402_COMC.870.990.V2.0.0.sdk200.DevKit #

```

4. MPP 适配

sample 代码：SS928V100_SDK_V2.0.2.2_MPP_Sample/src/megvii

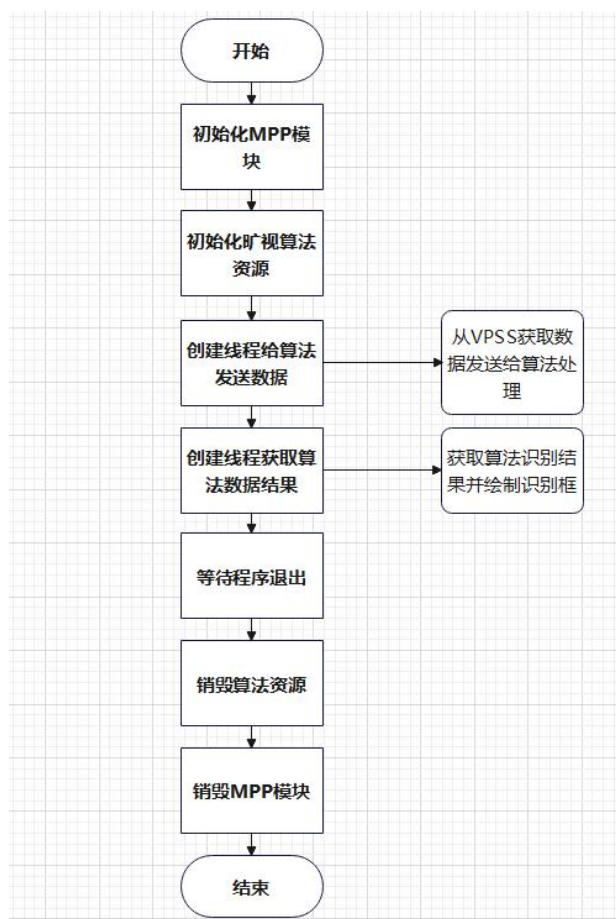
板端依赖库：

```
libmcv sdk_video.so  
libmegsearch_lite.so  
libss_mpi.so  
libss_ive.so  
libss_voice_engine.so  
libss_dnvqe.so  
libss_upvqe.so
```

Makefile 文件部分如下，参考 NPU sample 添加 NPU 库，同时加上旷视的库。

```
TARGET := sample_megvii  
  
MPI_LIBS += $(REL_LIB)/libss_ive.a  
MPI_LIBS += $(REL_LIB)/libss_md.a  
MPI_LIBS += $(REL_LIB)/libss_mau.a  
MPI_LIBS += $(REL_LIB)/libss_dpu_rect.a  
MPI_LIBS += $(REL_LIB)/libss_dpu_match.a  
MPI_LIBS += $(REL_LIB)/libss_dsp.a  
MPI_LIBS += $(REL_LIB)/svp_npu/libsvp_acl.a  
MPI_LIBS += -L$(REL_LIB)/svp_npu -lprotobuf-c  
  
CFLAGS += -I$(PWD)/../svp/common  
CFLAGS += -fstack-protector-strong  
CFLAGS += -L$(3RDPARTY_LIBS_PATH)  
CFLAGS += -O3  
CFLAGS += -L$(REL_LIB)/npu/stub -lascendcl  
  
MPI_LIBS += -lmcv sdk_video -lmegsearch_lite
```

MPP 流程图如下，主要新增线程发送数据给算法处理，以及创建线程获取算法识别结果绘制目标框。



5. 用户授权

因为通信认证文件 cert 包含了所有可授权的设备数量，实际开发过程中，可能不会将 cert 文件提供给客户（除非这个 cert 文件下的所有设备都是给该客户的），此时就需要由客户生成设备的指纹文件，然后由我们从旷视后台操作生成对应的授权文件给客户使用，操作如下图：



绑定授权

下载授权

通信认证信息

批量下载

设备SN 请输入

☐	设备SN	指纹文件	设备型号	设备授权上限	已授权数量	绑定时间	算法功能	原子有效期累计数量	操作
☐		详情	Hisi3402_COMC.870.990	1	1	2025-08-26 11:01:53	详情	1	下载授权文件
☐		详情	Hisi3402_COMC.870.990	1	1	2025-08-22 11:19:47	详情	1	下载授权文件
☐		详情	Hisi3402_COMC.870.990	1	1	2025-08-22 09:37:49	详情	1	下载授权文件
☐		详情	Hisi3402_COMC.870.990	1	1	2025-06-13 09:26:46	详情	1	下载授权文件



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